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Lesson 3: Advanced Multivariate Models and Practical Applications

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Practice Assignment: Advanced Multivariate Models and Practical Applications Quiz Submitted

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Module 3: Summative Assessment

Module 3 Summary

Your grade: 100%

Your best: 100% • Your highest score: 100%

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Advanced Multivariate Models and Practical Applications Quiz

Instructions

Review Learning Objectives

1. Suppose two instructors are being considered for one position. In the last class Instructor A taught, 15 out of the 19 students approved of the way the instructor managed the class. In the last class Instructor B taught, 20 out of the 27 students approved of the way the instructor managed the class. Using Bayesian inference, construct a 95% Credibility Interval for the difference in approval rates between the two classes. What might you conclude?

Help me practice

Let's chat

☒ The instructors are equivalent since the credibility interval includes 0.

☐ The instructors are equivalent since the credibility interval does not include 0.

☐ The instructors are different since the credibility interval includes 0.

☐ The instructors are different since the credibility interval does not include 0.

Correct

Correct

Nov 21, 3:36 PM CST

Unanswered

Retry

Your grade

To pass you need at least 80%. We keep your highest score.

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2. In a biology class, two instructors, Instructor A and Instructor B, are evaluated based on their recent classes. Instructor A received approval from 15 out of 19 students in the first class and 20 out of 27 students in the second class. Using Bayesian inference to estimate the approval rates, which of the following statements about Bayesian analysis is this correct to correct?

☒ The prior distributions for the approval rates of Instructor A and Instructor B are modeled using Beta distributions.

☐ The credibility interval for the difference in approval rates between Instructor A's two classes can be directly calculated using frequentist methods.

☐ The posterior distribution for the approval rates of Instructor A and Instructor B will be uniform distributions.

☐ The Bayesian approach assumes a normal distribution for the approval rates of Instructor A and Instructor B.

Correct

Correct

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3. In which step in our strategy for computation would conjugate distributions be considered for a Bayesian inference problem?

☒ Step 2

☐ Step 4

☐ Step 3

☐ Step 1

Correct

Correct

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